

Curriculum Vitae – Zirui Zhao

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EDUCATION	Doctor of Philosophy (Ph.D.) in Computer Science National University of Singapore	Aug 2020 - Present Advisor: Prof. Wee Sun Lee
	Bachelor of Engineering (B.Eng.) in Automation (EECS) Xi'an Jiaotong University	Aug 2016 - Jun 2020 GPA: 3.92/4.30
VISITING	Carnegie Mellon University , Pittsburgh, U.S.A. Summer Intern, Safe AI Lab	Jul 2019 - Sep 2019 Advisor: Prof. Ding Zhao
INTERESTS	Robotics; Robot Decision-making under uncertainty; Robot Planning with Large Language Models; Robot Perception; Human-robot interactions.	
SKILLS	Languages: Python, C++, MATLAB, $\LaTeX$. Application Tools: TensorFlow, PyTorch, Keras, Scikit, Numpy, Jupyter, ROS. Software: Carla, GitLab, GitHub.	
AWARDS	• Best Paper Runner-up LTAMP workshop @ Robotics: Science & Systems, 2023 • NUS Research Scholarship National University of Singapore, 2020 - 2024 • Excellent Graduate Xi'an Jiaotong University, 2020 • Dean's List Xi'an Jiaotong University, 2017 & 2018 & 2019 • 2nd price – 1989 Mechanical Alumni Scholarship Xi'an Jiaotong University, 2018 • Siyuan Merit Scholarship Xi'an Jiaotong University, 2017	
RESEARCH	[4] Z. Zhao, W. S. Lee, and D. Hsu. "Large Language Models as Commonsense Knowledge for Large-Scale Task Planning". In: <i>Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)</i>. 2023. [3] Z. Zhao, W. S. Lee, and D. Hsu. "Differentiable Parsing and Visual Grounding of Natural Language Instructions for Object Placement". In: <i>2023 IEEE International Conference on Robotics and Automation (ICRA)</i>. 2023, pp. 11546–11553. DOI: 10.1109/ICRA48891.2023.10160640. [2] R. Chen, W. Wang, Z. Zhao, and D. Zhao. "Active learning for risk-sensitive inverse reinforcement learning". In: <i>arXiv preprint arXiv:1909.07843</i> (2019). [1] Z. Zhao, Y. Mao, Y. Ding, P. Ren, and N. Zheng. "Visual-based semantic SLAM with landmarks for large-scale outdoor environment". In: <i>2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI)</i>. IEEE. 2019, pp. 149–154.	
TEACHING	CS3263 Foundations of AI Instructor: Prof. Tze-Yun Leong Tutor for course tutorials that cover Logic, Search, CSP, Bayesian Networks, Causal Networks, Markov Decision Process, and Reinforcement Learning.	Jan 2023 - Present National University of Singapore
REFERENCE	• Prof. Wee Sun Lee, Professor @ NUS. leews@comp.nus.edu.sg • Prof. David Hsu, Provost's Chair Professor @ NUS. dyhsu@comp.nus.edu.sg • Prof. Tze-Yun Leong, Professor @ NUS. leongty@nus.edu.sg • Prof. Ding Zhao, Associate Professor @ CMU. dingzhao@cmu.edu • Prof. Pengju Ren, Professor @ XJTU. pengjuren@xjtu.edu.cn	